STAC WARS: MUSTAFAR RESEARCH AGENCY

REPORT

- Rocket Specifications and Environment

- 6.88 kg

- aluminium body

- 1 kg payload

- Contrail J416-SP (booster)

- Cesaroni 567I125-10A (sustainer)

- 10 m/s wind at 270°

- the same stage separations and parachute events.

in two OpenRocket 6-DOF simulations. Everything was identical except the environment: Earth and Mustafar differ in environment in these parameters :

Parameter	Earth	Mustafar
Gravity	9.79 m/s²	6.50 m/s²
Surface pressure	101.3 kPa	75 kPa
Surface temperature	288 K	700 K
Air density	1.225 kg/m³	0.373 kg/m³

- RESULTS and Differences between simulations

Quantity	Earth	Mustafar	Change
Apogee	1922 m	3929 m	×2.04
Time to apogee	19.8 s	33.6 s	×1.70
Maximum velocity	206.7 m/s	261.9 m/s	27.00%
Maximum Mach number	0.626	0.473	-24.00%
Maximum acceleration	104.6 m/s²	107.9 m/s²	+3.3 m/s²
Peak dynamic pressure (½ρv²)	24.3 kPa	13.0 kPa	-46.00%
Stability margin, lowest / highest	1.97 / 4.58 cal	1.87 / 4.56 cal	about the same
Descent rate, payload container	3.65 m/s	5.32 m/s	×1.46
Descent rate, sustainer	3.61 m/s	5.25 m/s	×1.45
Downrange drift (container / sustainer)	4.10 / 4.16 km	5.87 / 5.97 km	×1.43
Flight time (container)	532 s	786 s	×1.48
Total speed at parachute deployment	40.9 m/s	39.2 m/s	similar

- PREDICTIONS and Tests

hypothesis : Lower gravity reduces weight and lengthens the climb

test : The rocket’s weight falls from 67.4 N to 44.7 N. We predicted a higher acceleration, less speed lost to gravity, and a longer coast. Maximum acceleration is 3.3 m/s² higher, matching the 3.3 m/s² drop in gravity. Over the 7.3 s powered phase, gravity removes about 72 m/s on Earth and about 48 m/s on Mustafar.

That difference of about 24 m/s is roughly a third of the 72 m/s by which the Mustafar burnout speed exceeds Earth's (243.6 vs 171.5 m/s vertical). Time to apogee rises from 19.8 s to 33.6 s. Held. The split between the gravity effect and the air effect is an estimate from the exported data, not an isolated run.

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hypothesis : Thinner air reduces drag.

test : Air density is 3.3 times lower. Because = P/(RT), pressure alone would cut the density by a factor of 0.74 and temperature alone by 0.41. Temperature therefore causes roughly three quarters of the density drop (in log terms). With the same thrust, the Mustafar rocket reaches a 27% higher maximum speed, while peak dynamic pressure is 46% lower. Held. The rest of the apogee gain comes from the drag reduction, which we could not separate from gravity in these two runs

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hypothesis : Thinner air raises parachute descent speed

test : For a fixed parachute, descent speed scales as $\sqrt{(g/\rho)}$. With Mustafar's 6.5 m/s² and ground density 0.373 kg/m³, we predicted a descent rate 1.48 times Earth's. The simulation gives 1.46 (container) and 1.45 (sustainer), within about 2% of the prediction. Held. Our parachutes were sized for Mustafar, and the same rocket would land at about 3.6 m/s on Earth, below the 4 m/s minimum.

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hypothesis : Thinner air changes aerodynamic stability.

test : The stability margin is essentially unchanged: the lowest value is 1.87 on Mustafar and 1.97 on Earth, and the highest are 4.56 and 4.58. The margin depends mainly on the geometry of the centre of gravity and centre of pressure. What changes is the strength of the aerodynamic forces. Peak dynamic pressure halves, and the flight path tilts less from vertical at 5 s (15.8° on Mustafar against 21.4° on Earth). Not held for the margin, held for the strength of the aerodynamic response

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hypothesis : High temperature raises the speed of sound

test : which lowers the Mach number. For air-like gas, a = $\sqrt{}$(γRT) gives about 340 m/s at the Earth surface and about 530 m/s at 700 K. The Mustafar rocket flies 27% faster but reaches a maximum Mach number of only 0.473, against 0.626 on Earth. Both flights stay subsonic. The simulated Mustafar air cools with altitude (to 529 K at 3.9 km), so the speed of sound falls on the way up but stays well above Earth's. Held.

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hypothesis: A 10 m/s side wind drives weathercocking and downrange drift

test : The wind was the same in both runs, so we could not isolate it. Drift tracks time aloft: average drift speed is 7.5 m/s on Mustafar and 7.7 m/s on Earth, about three quarters of the mean wind speed. Mustafar drifts 43% farther because it flies 48% longer. In both runs the rocket tilts 16-25° from vertical during the burn and carries about 40 m/s of sideways speed at apogee, which is consistent with weathercocking. Drift prediction held. Weathercocking is consistent with the data but was not isolated.

- CONCLUSION

Of the six predictions, five held (gravity, drag, descent rate, Mach number and drift).

The stability prediction held only for the strength of the aerodynamic response, not for the margin.

The Earth comparison also shows the design is tuned to Mustafar: the same rocket reaches only 1.9 km on Earth, and its parachutes would land too slowly there.